# Reproducibility Failures in Quantized Model Evaluation: Item Ordering Moves 4-Bit Accuracy and Leaves `bf16` Exactly Unchanged

**Anonymous ACL submission**

## Abstract

Reported quantization degradation is a difference between two measurements, and a difference inherits every property of the instrument that produced it. We evaluate three instruction-tuned models at five precisions on reasoning benchmarks, holding the engine, decoding configuration and prompts fixed, and vary properties of the evaluation apparatus that no protocol we are aware of reports. Submitting the same items in a different order changes 4-bit accuracy by up to 1.09 points of standard deviation across 10 orderings while leaving `bf16` accuracy *exactly* unchanged; disabling the quantized matrix-multiply kernels drives the variation to zero, so the mechanism is established by intervention. Generation is bitwise reproducible *within* a process, with zero differing items across 42,380 paired comparisons, and is *not* reproducible across processes, where the same configuration diverges on up to 45% of items; the common practice of repeating a measurement in one process therefore cannot detect it. Under a fixed generation budget, arms differ by an order of magnitude in how many tokens their scored answer discards. We further show that the scoring rung interacts with precision *heterogeneously* across tasks and quantization methods, established by pre-registered contrasts, so no single protocol-level correction restores comparability. We release 105,950 generations with per-item termination and length metadata, and an audit of our own extraction pipeline, which had silently degraded 2% of that corpus until a census found it.

## 1 Introduction

Reported quantization degradation is a difference between two measurements, and a difference inherits every property of the instrument that produced it. This paper asks which properties of the evaluation apparatus move that difference, and answers with three that are not reported by any protocol we are aware of.

The framing matters because part of the answer is already known. Kurtic et al. (2025) showed that reported degradation had been overstated and attributed the excess to untuned quantization-side hyperparameters. We treat that as established and prior. Our concern is the residual that survives quantization-side tuning, and our mechanisms are orthogonal to theirs: they act on the evaluation, not on the quantized model.

**An artifact in a level is not an artifact in a difference.** That distinction organises the paper. It is well known that lenient answer extraction raises absolute scores (Jo et al., 2025; Yu et al., 2025), and that prompt formatting moves them substantially (Sclar et al., 2024). What has not been established is whether such choices change the quantization *delta*, which is the quantity papers actually report. A factor that shifts every arm equally is a nuisance; a factor that shifts arms *differently* is a confound in every published comparison. We test for the second, and find it.

**Three protocol factors, one of which we can turn off.** Serving the same items through the same engine and varying only their submission order changes 4-bit accuracy by up to 1.09 points of standard deviation across 10 orderings, while leaving `bf16` accuracy *exactly* unchanged at both scoring rungs. Disabling the quantized matrix-multiply kernels drives that variation to zero on 2 arms, so the mechanism is established by intervention rather than inferred. Separately, generation is bitwise reproducible *within* a job and is not reproducible *across* jobs, which means the standard remedy of repeating a measurement in one process cannot detect it. And under a fixed generation budget, arms that emit longer traces lose answers to truncation at differing rates, converting a length difference into measured degradation.

**The effect reaches the reported quantity, and not uniformly.** The scoring rung interacts with precision, and the interaction is *heterogeneous*: it differs across tasks and across quantization methods, both established by pre-registered contrasts rather than observed. The consequence is the practical one. If the interaction were uniform, a single correction would remove it. Because it is not, no protocol-level adjustment recovers a comparable number, and the only remedy is to report the choices.

**This is not an argument for scoring more permissively.** Two facts prevent it. Ren et al. (2026) find that 8.2–44.1% of answers graded correct are reached by shortcut hacking, so leniency overcredits. And we measure that lenient extraction also *destroys* correct answers, at a small but nonzero rate, so it is not a monotone relaxation of strict extraction but a different selection rule with two-sided error. The contribution is to *calibrate* extraction, not to loosen it.

**The literature's centre of gravity runs against us, and that belongs here.** The dominant finding in this area is that benchmarks *understate* quantization damage: Marchisio et al. (2024) report a 1.7% automatic drop in Japanese corresponding to 16.0% by human judgement, and Dutta et al. (2024) show that models matched on accuracy still diverge behaviourally. We regard both directions as simultaneously true, because they are different quantities. An accuracy delta can over-penalise formatting *and* under-detect behavioural change. Nothing here licenses a conclusion that quantization is benign; the claim is about the comparability of reported numbers, not about deployment risk.

**Contributions.** (i) Item ordering moves 4-bit accuracy and leaves `bf16` exactly unchanged, with the mechanism pinned to a named kernel by intervention, and four coverage gaps identified in a batch-invariant serving mode. (ii) Generation is reproducible within a job and not across jobs, so reproducibility must be measured across job boundaries; we report four candidate mechanisms we refuted and the measurements that refuted them. (iii) The precision-by-scoring interaction is real and heterogeneous across tasks and methods, so no single correction recovers comparability. (iv) A corpus of 105,950 generations with per-item termination and length metadata, released so that any extractor can be re-evaluated without a GPU. (v) An audit of our own extraction pipeline, which had silently degraded 2% of that corpus until a census found it, offered as an instance of the effect we measure rather than as an aside.

## 2 Related Work

**A live contradiction about generation length.** Whether low-bit quantization lengthens chain-of-thought output is currently disputed. Lian et al. (2026) report that INT4/INT3 quantization can preserve accuracy while *increasing* reasoning-token usage, and propose a CoT Token Inflation Ratio to quantify it; Lotfi et al. (2026) likewise find that aggressive post-training quantization lengthens CoT. Liu et al. (2025) report the opposite: "contrary to expectations, quantized models do not exhibit increased output lengths." We take no position on which generator behaviour is correct, because the disagreement has a consequence that neither side measures. Under a *fixed* generation cap, a model that emits longer traces is scored wrong more often whenever its answer falls beyond the cap — a scoring outcome, not a competence one. No study on either side of this dispute sweeps the generation cap, so no study can separate the two. That separation is our H1.

**Degradation has already been partly explained — by other causes.** Kurtic et al. (2025) show that reported quantization degradation was overstated, and attribute the excess to untuned quantization-side hyperparameters. We treat that result as established and *prior*, not as a target: our claim concerns the residual that remains after quantization-side tuning, and our mechanism is orthogonal to theirs. Lotfi et al. (2026) supply the most direct antecedent for the present work, finding that in as many as 52% of quantized-model failures the model reaches the correct answer in an intermediate step but does not emit it as the final answer. Their interest is in fixing the generator, via a logit penalty on overthinking markers; ours is in what such failures do to a reported accuracy delta when they are left in place.

**The centre of gravity runs the other way.** The dominant finding in this literature is that benchmarks *understate* quantization damage, not overstate it. Marchisio et al. (2024) find a 1.7% automatic drop in Japanese corresponds to 16.0% by human judgement, and Dutta et al. (2024) show that models matched on accuracy still diverge be-

haviourally — the "flips" framing, which is theirs and which we do not claim. We regard both directions as simultaneously true: an accuracy delta can over-penalise formatting *and* under-detect behavioural change, because these are different quantities. Our contribution is to the first, and it does not license any conclusion about the second.

**Calibrating extraction, not loosening it.** An argument that evaluation should simply be more permissive is already refuted. Ren et al. (2026) find that 8.2–44.1% of answers graded correct are reached by shortcut hacking, so lenient extraction over-credits as surely as strict extraction under-credits. We therefore treat the strict/permissive contrast as a measurement instrument to be characterised rather than a knob to be turned in one direction, and we report both rungs throughout.

**Prompt- and protocol-sensitivity.** Hua et al. (2025) ask whether prompt sensitivity is a genuine model property or an artifact of evaluation, and we are structurally the sequel: they vary the prompt, we vary the scorer, and both ask whether a reported effect survives the instrument. Their null result on MATH deserves emphasis rather than a footnote — extraction there already uses symbolic equivalence, so a well-engineered verifier leaves little for a protocol effect to explain. This bounds our claim rather than supporting it, and it is why our task selection targets benchmarks whose canonical extractors are regular expressions.

**The nearest prior work, and what it deliberately designed out.** Pape et al. (2026) is the closest study to our §4.1 and must be distinguished precisely. They survey 35,000 publications, find that only 36% of papers without released code name even their inference backend, and show that switching backend alone moves GSM8K by 16.6 points. We adopt their framing and their survey rather than repeating it. But their controlled comparison runs at *batch size one*, chosen explicitly to prevent batching-induced instability, holds precision at FP16, and varies only random seeds within an engine; batch invariance and quantized kernel selection do not enter. The instability they design out is the one we make the object of study: we hold the engine, the weights and the decoding configuration fixed, vary only which items share a batch, and attribute the resulting divergence to a named quantized kernel by intervention. Their root-cause analysis independently supports our wider point — their two largest outliers were a spurious BOS injection and an unnoticed repetition penalty, worth 8.3 and 11.7 points, both configuration defaults rather than model properties.

Sclar et al. (2024) establish that spurious formatting choices move scores substantially, and Jo et al. (2025) and Yu et al. (2025) show that answer extraction from reasoning traces is itself an unreliable step. Schaeffer et al. (2023) provide the methodological template we follow: a headline effect reattributed, in part, to the measurement instrument. The lm-evaluation-harness (Biderman et al., 2024) is our instrument, and its canonical filters are the object of study rather than an implementation detail.

**Serving configuration is a confound we pin, not an axis we explore.** Pape et al. (2026) find that inference-backend choice alone moves benchmark scores by up to 16.6 points, and Masoudian et al. (2026) attribute roughly 39% of out-of-the-box variance to it. We read this as a reason to freeze the engine, version, and every decoding parameter across all arms, which we do and report; a floating serving configuration would make precision effects unidentifiable. We claim no novelty on this axis.

**Adjacent evaluations.** Broad compression benchmarks (von Rad et al., 2026; Wang et al., 2025) and task-difficulty studies (Lee et al., 2025) establish the accuracy landscape we work inside; Lee et al. (2025) in particular report that quantization disrupts instruction-following, which is the strongest available objection to our framing and which our compliance/competence split is designed to address directly rather than deflect. Prasad and Pal (2026) run a factorial quantization analysis, so factorial design is not our contribution; their outcome domain is safety alignment under sampling temperature, which is disjoint from ours. Kurt (2026) report the observation this paper sets out to explain — strict-versus-flexible GSM8K gaps of 41 to 64 points across fourteen `llama.cpp` quantizations — without interpreting the gap; we reproduce their numbers as an external anchor. Oladri et al. (2026) and Rasmussen and Qin (2026) report related failure-mode and scoring-rule effects, the former over FP32/FP16/NF4 only and so without bearing on GPTQ or AWQ.

**What is new here.** Permissive extraction is well known to raise absolute scores. What has not been established is whether it changes the quantization *difference*: an artifact in a level need not be an artifact in a delta, and the two are routinely conflated. We decompose the measured strict delta into competence, compliance, and truncation components, pre-register the decomposition before any generation exists, and report the residual precision effect that survives it.

## 3 Method

### 3.1 Experimental design

We evaluate three instruction-tuned models spanning a 3–8B size gradient: Qwen2.5-3B-Instruct, Phi-3.5-mini-instruct, and Llama-3.1-8B-Instruct. Each is pinned by commit SHA rather than tag, since tags move and reproducibility holds only within identical versions.

Five precision arms form a dose–response rather than a two-point comparison: `bf16`, GPTQ-8bit, GPTQ-4bit, AWQ-4bit, and bitsandbytes NF4. A monotone bf16 $\rightarrow$ 8-bit $\rightarrow$ 4-bit trend is a substantially stronger internal-validity argument than a single contrast, because a protocol artifact has no reason to track bit-width monotonically.

All manipulations are one-factor-at-a-time from a single fixed reference condition, so every reported effect is attributable to the one factor varied. NF4 is included in the reference condition but excluded from the token-budget sweep, and we state the reason rather than the exclusion alone: NF4 has no CUDA kernels in our serving engine, and it is quantized *in flight* at load rather than loaded from a pre-made checkpoint. That is a backend difference layered on the bit-width difference, which we name explicitly rather than absorb.

### 3.2 Quantization and calibration

This is the most likely attack surface on the paper, so we state it in full. Both GPTQ and AWQ are produced by one tool, through one calibration code path, at *one* normalised budget: 512 sequences of at most 2048 tokens drawn from a single pinned shuffle of `ultrachat_200k`, identical for every arm, every bit-width, and every model. We report the realised token count rather than the product of those two numbers, since sequences shorter than the cap contribute their own length.

Normalisation is necessary because the reference implementation's own examples differ by **8**$\times$ in calibration budget between the two methods (GPTQ at 512$\times$2048 symmetric; AWQ at 256$\times$512 asymmetric). Accepting those defaults would mean a GPTQ-versus-AWQ difference confounded with an 8$\times$ calibration difference *and* a symmetry difference. We therefore normalise AWQ to symmetric 4-bit weight-only quantization, which is a deliberate deviation from its published example, and we flag it as such.

Group size is 128 and activation reordering is disabled for every GPTQ arm, matching the vendor's own released checkpoints. Disabling reordering is set explicitly rather than left to the library, whose default silently applies a static reordering. Every parameter above is hashed into a configuration digest carried on each generation record, so a reader can verify that no arm drifted.

All arms are self-quantized with identical settings. Vendor checkpoints are used only as an external cross-check, never mixed into the main matrix, because a widely used third-party GPTQ checkpoint for one of our models enables activation reordering against the vendor's own setting — exactly the kind of silent confound this design exists to exclude.

### 3.3 The scoring ladder

Every generation is stored once and scored many times. We apply a ladder of extractors to the *same* stored text, so the entire ladder costs zero additional generation and every rung sees byte-identical inputs:

1. **strict** — the evaluation harness's canonical regular expression, used verbatim.
2. **strict-last** — the same expression selecting the *last* match instead of the first, which separates answer *position* from extraction *strictness*.
3. **permissive** — the harness's flexible extractor, with its rules enumerated in the pre-registration.
4. **symbolic** — equivalence checking rather than string matching.
5. **judge** — a trained answer verifier (Chen et al., 2025).

**The headline contrast is rung 1 versus rung 2.** Both are deterministic, auditable, and judge-independent, which is what makes judge validation a robustness check rather than a load-bearing

dependency: if the judge fails its agreement gate, the paper's main claim is unaffected. Rung 1b exists because a harness that takes the first match will score a model that restates its answer differently from one that does not, independent of correctness.

**An audit of our own extractor, and what it found.** Because this paper's claim is that unreported measurement choices move reported numbers, we audited our own extraction rather than assuming it. Our permissive extractor applied a commitment-phrase rule ahead of the harness's flexible-extract regex, and returned the committed answer with its unit attached (`"5 days"` against a gold of `"5"`). Because the harness's regex admits no alphabetic character, it could not have produced those values: the defect was ours, ordered ahead of published practice, and it returned worse answers than the harness would have on 2,068 of 105,950 generations. A full census of every rung-monotonicity violation is what found it; our unit tests passed throughout, because the fixture set contained no generation that continued past its committed answer. Two interaction estimates changed conclusion when it was repaired and one changed sign. All numbers reported here are post-repair. We record the episode because it is an instance of the effect this paper measures rather than an aside: a well-intentioned extractor improvement silently degraded 2% of a six-figure corpus, our unit tests passed throughout, and only an exhaustive audit surfaced it. It is also why we report a census of every affected item rather than a sampled estimate.

### 3.4 Compliance, competence, and truncation

We decompose the measured strict delta into three additive components: a *competence* component (the model does not know the answer), a *compliance* component (it knows but does not emit the answer in the expected form), and a *truncation* component (it would have emitted the answer but ran out of budget). Recovering the third requires two fields that cannot be reconstructed after the fact — the engine's termination reason and the generated token count — so both are recorded on every generation, and our writer refuses any record missing either.

Compliance is treated as a dependent variable in its own right, not merely as a correction term. We also distinguish an answer being *extractable* from the model having *committed* to it: a fallback rule that returns the last number in a truncated trace has extracted something the model never asserted, and counting that as compliance would inflate the very component we are trying to measure.

### 3.5 Serving configuration and determinism

The serving configuration is frozen and identical across every arm: one engine, one pinned version, one set of decoding parameters, greedy decoding throughout. This is not incidental rigour. Backend choice alone moves benchmark scores by up to 16.6 points (Pape et al., 2026) and accounts for roughly 39% of out-of-the-box variance (Masoudian et al., 2026), so a serving configuration that floated across arms would make precision effects unidentifiable regardless of sample size.

Greedy decoding is not by itself sufficient for reproducibility, because inference engines batch requests and batched reductions are not order-invariant. We therefore enable batch-invariant kernels, run with tensor parallelism of one (batch invariance is broken above one), disable prefix caching (it changes greedy output and is explicitly outside the invariance guarantee, and it would otherwise help throughput considerably on our shared few-shot prefixes), and fix the maximum number of co-resident sequences to a value *measured* rather than assumed: the largest setting that is bitwise reproducible across repeated identical runs for *every* precision arm, which is a minimum over arms because a single shared value is required.

We report that value, the per-arm bitwise disagreement rates behind it, the GPU model, and the CUDA and driver versions. Where a quantized arm is less reproducible than `bf16`, the difference is a noise floor on every contrast involving that arm, and we pre-commit to claiming no effect smaller than it.

## 4 Results

### 4.1 Evaluation-set ordering (H8)

We serve every arm through a single engine version (1.0.0) with batch-invariant kernels enabled, tensor parallelism disabled, prefix caching off, and a fixed decoding configuration, then vary one parameter that no evaluation protocol we are aware of reports: *the order in which items are submitted*. Because requests are batched dynamically, an item's position determines which other items share its batch, and therefore the shapes over which the

| | strict | | permissive | |
|---|---|---|---|---|
| arm | SD | range | SD | range |
| BF16 | 0.00 | 0.00 | 0.00 | 0.00 |
| GPTQ W4A16 | 0.87 | 2.80 | 0.75 | 2.40 |
| AWQ W4A16 | 0.52 | 2.00 | 1.09 | 4.20 |
| bnb NF4 | 0.11 | 0.40 | 0.10 | 0.20 |

Table 1: Accuracy across 10 orderings of the same 500 items, in accuracy points, measured at the served batch size $B = 32$ (§3.5). The standard deviation is the inferential quantity; the range is descriptive only, being unstable and monotone in the number of orderings sampled. `bf16` is identical under every ordering at both rungs.

| | text divergence | | accuracy SD | |
|---|---|---|---|---|
| arm | Marlin on | off | Marlin on | off |
| GPTQ W4A16 | 53.2 | 0.0 | 0.42 | 0.00 |
| AWQ W4A16 | 53.6 | 0.0 | 0.35 | 0.00 |
| bnb NF4 | 49.4 | 49.4 | 0.31 | 0.31 |

Table 2: Paired intervention at $B = 8$: identical inputs, identical engine, with and without the Marlin and Machete kernels. Text divergence and accuracy SD in percentage points. bnb NF4 is unaffected, indicating a second and distinct mechanism.

engine's reductions are performed.

Holding the ordering fixed, generation is exactly reproducible: the repetition floor is 0.000 % in all 15 cells. So every difference reported below is attributable to ordering alone, not to run-to-run variation.

**Aggregate accuracy moves, and only for quantized arms.** Table 1 gives the standard deviation of accuracy across 10 orderings of the same 500 items at the served batch size we adopt. For `bf16` the accuracy is *identical* under all 10 orderings, at both scoring rungs: 2 of 2 cells show exactly zero variation. Every quantized arm moves, by up to 1.09 points of standard deviation.

**Why we report a standard deviation and not a spread.** A minimum–maximum spread is monotone in the number of orderings sampled, so it grows with effort rather than converging. We can show this rather than assert it: the same quantity measured over three orderings gave a spread of 1.6 points and over 10 gives 4.2, with no change to the underlying effect. Any spread reported over a handful of runs is therefore a lower bound on the spread its own protocol would have produced with more. The standard deviation, by contrast, stayed within a narrow band across arms and rungs.

Three properties matter more than the magnitude. The shift is **deterministic** given an ordering rather than sampling noise, so it cannot be averaged away by repeated runs. It is **bidirectional** — accuracy moves both up and down across orderings, so no ordering is the correct one and the effect cannot be calibrated away. And it is **attributable by intervention** rather than inferred, as follows.

**The mechanism is a named kernel.** Disabling Marlin and Machete through `VLLM_DISABLED_KERNELS`, changing nothing else, drives order-induced divergence to zero on 2 arms (Table 2) — on generated text *and* on accuracy. These are the quantized matrix-multiply kernels vLLM selects by default, and they are not covered by its batch-invariant mode: the AWQ kernel path carries a batch-invariance guard, the GPTQ path carries none, and the sampler and kernel autotuning paths are likewise unguarded. We report this as a defect in the feature rather than in either kernel; AWQ has the guard and is nonetheless affected, so the guard as implemented is insufficient.

Two controls rule out the obvious alternatives. Enabling kernel autotuning leaves every figure bit-identical, so the effect is not an autotuning artifact. And bnb NF4 is unmoved by the intervention while remaining order-sensitive, so at least one further mechanism exists that is Marlin-independent; we note that it becomes *less* order-sensitive at larger batch sizes, the reverse of every other arm, and leave the cause unresolved.[1]

**The frontier.** Table 3 reports divergence at all three measurement levels and all three batch sizes, rather than only at the configuration we adopt. The three levels differ by roughly an order of magnitude: generated text diverges far more often than the extracted answer, which diverges more often than scored correctness. Reporting text divergence alone would substantially overstate the impact on any reported number, and reporting correctness alone would conceal how much of the trace is un-

[1] `bitsandbytes` is proposed for out-of-tree maintenance in vLLM RFC #39583, on performance grounds. Two explanations are consistent with our data — a data-dependent reduction whose result concentrates at larger batch sizes, or a different kernel-selection path entirely — and we do not have evidence to separate them.

| | $B = 8$ | | | $B = 32$ | | | $B = 64$ | | |
|---|---|---|---|---|---|---|---|---|---|
| arm | text | str. | perm. | text | str. | perm. | text | str. | perm. |
| BF16 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| GPTQ W8A16 | 50.6 | 5.0 | 6.8 | 53.4 | 5.8 | 7.0 | 51.4 | 5.6 | 7.8 |
| GPTQ W4A16 | 53.2 | 5.4 | 5.8 | 59.0 | 7.4 | 8.2 | 59.4 | 7.4 | 11.2 |
| AWQ W4A16 | 53.6 | 4.2 | 7.6 | 54.8 | 5.6 | 9.4 | 54.8 | 5.2 | 9.6 |
| bnb NF4 | 49.4 | 7.6 | 6.6 | 5.2 | 0.0 | 0.2 | 1.8 | 0.0 | 0.0 |

Table 3: Order-induced divergence at three measurement levels — generated text, and correctness under strict and permissive extraction — by served batch size $B$. The frontier, not the single configuration we adopt, is the result: it bounds what a third party can reproduce.

stable.

Notably the permissive rung is the more order-sensitive of the two at the answer level, because it accepts the last number anywhere in the trace and the trace is what diverges, whereas strict extraction requires a committed marker that is comparatively stable. Leniency therefore buys recovery of truncated answers at the cost of exposure to batch-composition noise — which is an argument for calibrating extraction rather than loosening it (§3.3).

**Scope.** [**TODO:** H8 scope — restate once the 10-ordering confirmatory run (pre-registered, Appendix C) lands: SD over 10 orderings, 4 arms at $B = 32$, with the CORRECTED terminators, so it is not directly comparable to the 3-ordering figures above and supersedes rather than extends them. Current SD is weakly estimated at 3 orderings. Also add the one comparative sentence — ordering spread as a fraction of the bit-width effect measured on the SAME items — which requires the reference runs and MUST NOT be sourced from published deltas on other models and tasks, which §2 argues are mutually incomparable.]

### 4.2 Truncation (H1)

[**TODO:** H1 — degradation vs generation budget; mean/median n_gen_tokens by precision with CIs, settling 2606.25519 vs 2504.04823.]

### 4.3 Recoverable failures (H2)

[**TODO:** H2 — gold-in-trace rate, permissive and conservative variants, always reported net of the incidental-match false-positive floor.]

### 4.4 Precision × scoring interaction (H3)

[**TODO:** H3 — GLMM interaction as an odds ratio and as a marginal effect in accuracy points with 95% CI, plus paired McNemar. A NULL IS PUBLISHABLE and must not read as failure.]

### 4.5 Decomposition (H4)

[**TODO:** H4 — THE CENTREPIECE TABLE. Design it first and let it drive what gets logged. Always-compliant stratum size as a first-class number; compliance-transition matrix; Manski bounds.]

### 4.6 Prompt format (H5) and the structural control (H6)

[**TODO:** H5 zero-shot arm; H6 ARC-Challenge under log-likelihood vs generative scoring on the same 500 items — the gap between modes IS the extraction cost.]

### 4.7 Statistical resolution of published deltas (H7)

[**TODO:** H7 — scrupulously neutral: "many papers report aggregate deltas without a paired test", NEVER "the deltas are noise".]

## 5 Discussion

**What a reader should change, and it is not a threshold.** Our results do not license a rule of the form "report accuracy to one decimal place" or "average over $k$ runs". Averaging cannot help with a factor that is deterministic given a configuration, and precision is not the problem. What the results license is narrower and cheaper: three configuration choices currently go unreported, each of them moves the quantized arms and not the `bf16` baseline, and reporting them costs nothing. Item submission order, whether the compared arms were generated in the same process, and the generation budget with its truncation rate are the three. A protocol that reports them is comparable to another protocol that reports them; one that does not is not comparable to anything, including itself on a later day.

**Why repeating a measurement does not detect this.** The natural defence against measurement noise is to run it twice, and here that defence fails in a specific and instructive way. Within a single process, generation is bitwise reproducible: we observe zero differing items across 42,380 paired comparisons. Across process boundaries, the same configuration on the same hardware diverges on up to 45% of items for one model, and moves its accuracy by over a point. So a

practitioner who repeats a measurement in one process obtains perfect agreement and concludes the pipeline is deterministic, which is exactly the wrong conclusion, reached by exactly the check a careful person would run. Reproducibility has to be assessed across job boundaries, and we know of no protocol that does so.

**The mechanism we did not find, and what we ruled out.** We report the cross-process effect without a mechanism, and we state what was eliminated so that the gap is bounded rather than open. Four candidates were tested and refuted. Differing KV-cache block counts from per-invocation memory profiling: refuted, because the engine-reported cache sizes are byte-identical between jobs, cell for cell. Batching scheduled by memory rather than by sequence count: refuted, because one arm is stable at a concurrency of $12.36\times$ while another diverges at $10.70\times$, so the two are not separable. Weight provenance, base repository against local checkpoint: refuted, because it predicts one model exactly and is contradicted by the other two. Sampling configuration, including a hidden repetition penalty read from each model's own generation config: refuted, because the model shipping the most aggressive defaults is the only one that is perfectly reproducible. What remains untested is state that is compiled or cached once per process, which is consistent with the within-process zero, and we identify it as the next hypothesis rather than claiming it.

**Heterogeneity is the result that matters practically.** Had the scoring interaction been uniform across arms and tasks, it would be a nuisance parameter with a correction. It is not: it differs across tasks with the arm held fixed, and across quantization methods with the task held fixed, both by preregistered contrasts. A single adjustment therefore cannot restore comparability, and the sign of the effect depends on which method and which task a paper happened to use. That is why we frame the contribution as reporting rather than correcting.

**On auditing one's own instrument.** The largest single error we found was our own. An extraction rule we had added ahead of the harness's returned worse answers than the harness would have, on 2,068 of 105,950 generations, and our unit tests passed throughout because no fixture exercised a generation that continued past its committed answer. Two of our interaction estimates changed conclusion when it was repaired and one changed sign. We report this because it is the same phenomenon as the rest of the paper observed from the inside, and because it suggests that a census over every item is worth more than a fixture set when the quantity of interest is a small differential.

## 6 Conclusion

Reported quantization degradation is a difference between two measurements, and we show that three unreported properties of the measurement apparatus move that difference. Item submission order changes 4-bit accuracy while leaving `bf16` accuracy exactly unchanged, with the mechanism established by disabling a named kernel. Generation is bitwise reproducible within a process and not across processes, so the standard practice of repeating a run in one process cannot detect the variation that matters. And a fixed generation budget converts a difference in trace length into measured degradation at rates that differ by arm. The scoring rung interacts with precision heterogeneously across tasks and methods, so no single correction recovers comparability, and the remedy is to report the choices rather than to adjust for them. We release the full generation corpus so that any extraction rule can be re-evaluated without a GPU, including ours, which we found to be defective by auditing it.

## Limitations

**The reproducibility claims are single-hardware and single-engine.** Every determinism result holds for one GPU (L40S, SM 8.9), one engine version, and one CUDA and driver pair, all of which we report. Batch invariance is version-specific and kernel-specific, so neither the ordering result nor the cross-job result should be read as a claim about other hardware. We also cannot serve a stock AWQ checkpoint in `bfloat16` on this engine version at all, which is itself a comparability limitation rather than only an inconvenience: published AWQ numbers are not dtype-comparable to GPTQ unless dtype is reported.

**The cross-job mechanism is undiagnosed.** We establish that generation is bitwise reproducible within a job and not across jobs, and we report four refuted hypotheses with the measurements that refuted them. We do not identify the cause. The finding is therefore an observation about what a third party will encounter, not an explanation, and

our recommendation is correspondingly procedural rather than a fix.

**The competence decomposition rests largely on bounds.** The always-compliant stratum falls below half the paired items in 14 of 27 cells, and in the worst case (Qwen on the hash-marker GSM8K arm) it is three items. We therefore report the point estimate and the Manski bounds together for every cell, with the stratum size as a first-class number, rather than presenting a point estimate whose subgroup is too small to support it. Wide bounds are reported as wide. The decomposition is a supporting analysis in this paper, not a headline.

**Scope deliberately excluded, and what that costs.** There is no factual-recall task: TriviaQA was cut, so every task here is a reasoning task whose canonical extractor is a regular expression, and the structural control is the ARC mode pairing rather than a second domain. Our main matrix is self-quantized throughout, at a normalised calibration budget, which isolates the recipe but means we do not speak to arbitrary published checkpoints. Three models cannot support a variance estimate, so model is a fixed effect and no claim generalises to a model population. Finally, our anchor comparison against official checkpoints tests comparable quality at matched settings rather than reproduction, because the official checkpoints were calibrated on data we do not have; one of its three arms failed its pre-registered band, and we report the failure and the grounds on which we proceeded.

## Ethics Statement

This work evaluates existing open-weight models on existing public benchmarks and introduces no new model, no new dataset of human origin, and no deployment. We see three ethical considerations worth stating.

**Licensing, and one model we can run but not redistribute.** Every model and dataset is used within its licence. Two cases need naming rather than assuming. Qwen2.5-3B-Instruct is released under the Qwen research licence rather than Apache-2.0, and we report it as such. Our primary judge is released under CC-BY-NC-ND-4.0, which permits us to run it and report numbers but forbids redistributing the weights or any derivative, so the artifact release contains our judging code and its outputs and not the model. We name a second, Apache-2.0 judge for exactly this reason, so that the reliability claim can be reproduced by someone unwilling to accept the first licence.

**Annotation was done by the authors, unpaid, and that has a cost.** The human labels used to validate the judge were produced by the authors rather than by recruited annotators. No payment question therefore arises, but neither does the independence that recruited annotation would provide: the annotators knew the study's hypotheses, which is why the condition column is omitted from the annotation file entirely rather than merely hidden, why shuffling is independent per annotator with published seeds, and why the adjudication rule was fixed in writing before any disagreement was seen. We report inter-annotator agreement alongside judge agreement precisely so that a reader can see the ceiling our own labels impose.

**What a paper about measurement error could be misused to argue.** Our findings show that reported quantization degradation is in part an artifact of evaluation protocol. That could be read as licence to dismiss reported degradation generally, and we do not support that reading. The dominant finding in this literature runs the other way, with automatic metrics *understating* damage relative to human judgement, and our own results are about the *difference* between arms under a fixed protocol rather than about whether quantization is safe to deploy. We also report that leniency in extraction has two-sided error, so nothing here argues for scoring more permissively as a general remedy. The actionable content is that unreported measurement choices should be reported, which raises the cost of a claim rather than lowering it.

Compute is reported in full: every run, its duration and its cost are recorded in the released log, totalling roughly 35 GPU-hours on a single accelerator.

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## References

Stella Biderman, Hailey Schoelkopf, Lintang Sutawika, Leo Gao, Jonathan Tow, Baber Abbasi, Alham Fikri Aji, Pawan Sasanka Ammanamanchi, Sidney Black, Jordan Clive, Anthony DiPofi, Julen Etxaniz, Benjamin Fattori, Jessica Zosa Forde, Charles Foster, Jeffrey Hsu, Mimansa Jaiswal, Wilson Y. Lee, Haonan Li, and 11 others. 2024. Lessons from the trenches on reproducible evaluation of language models. *arXiv preprint arXiv:2405.14782*.

Ding Chen, Qingchen Yu, Pengyuan Wang, Mengting Hu, Wentao Zhang, Zhengren Wang, Bo Tang, Feiyu Xiong, Xinchi Li, Chao Wang, Minchuan Yang, and Zhiyu Li. 2025. xverify: Efficient answer verifier for reasoning model evaluations. *arXiv preprint arXiv:2504.10481*.

Abhinav Dutta, Sanjeev Krishnan, Nipun Kwatra, and Ramachandran Ramjee. 2024. Accuracy is not all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Andong Hua, Kenan Tang, Chenhe Gu, Jindong Gu, Eric Wong, and Yao Qin. 2025. Flaw or artifact? rethinking prompt sensitivity in evaluating llms. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Hwiyeol Jo, Joosung Lee, Jaehone Lee, Sang-Woo Lee, Joonsuk Park, and Kang Min Yoo. 2025. Finding answers in thought matters: Revisiting evaluation on large language models with reasoning. *arXiv preprint arXiv:2510.14773*.

Uygar Kurt. 2026. Which quantization should i use? a unified evaluation of llama.cpp quantization on llama-3.1-8b-instruct. *arXiv preprint arXiv:2601.14277*.

Eldar Kurtic, Alexandre Marques, Shubhra Pandit, Mark Kurtz, and Dan Alistarh. 2025. "give me bf16 or give me death"? accuracy-performance trade-offs in llm quantization. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*.

Jemin Lee, Sihyeong Park, Jinse Kwon, Jihun Oh, and Yongin Kwon. 2025. Exploring the trade-offs: Quantization methods, task difficulty, and model size in large language models from edge to giant. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence (IJCAI)*.

Xinyu Lian, Walid Krichene, Beichen Huang, Masahiro Tanaka, Olatunji Ruwase, Li Zhang, and Minjia Zhang. 2026. Quantization inflates reasoning: Token inflation as a hidden cost of low-bit reasoning models. *arXiv preprint arXiv:2606.25519*.

Ruikang Liu, Yuxuan Sun, Manyi Zhang, Haoli Bai, Xianzhi Yu, Tiezheng Yu, Chun Yuan, and Lu Hou. 2025. Quantization hurts reasoning? an empirical study on quantized reasoning models. In *Conference on Language Modeling (COLM)*.

Sanae Lotfi, Polina Kirichenko, Steven Li, and Zechun Liu. 2026. Quantized reasoning models think they need to think longer, but they do not. *arXiv preprint arXiv:2606.00206*.

Kelly Marchisio, Saurabh Dash, Hongyu Chen, Dennis Aumiller, Ahmet Üstün, Sara Hooker, and Sebastian Ruder. 2024. How does quantization affect multilingual llms? In *Findings of the Association for Computational Linguistics: EMNLP 2024*.

Shahed Masoudian, Passant Shafaei, Monorama Swain, and Markus Schedl. 2026. What we observe as llm behavior can be a side-effect of inference backend. *arXiv preprint arXiv:2608.04714*.

Renuka Oladri, Mohan Vamsi Varadaraju Priya, and Jerry Wu. 2026. Silent failures in quantized llm reasoning: A taxonomy-based analysis of hollow convergence and failure mode shifts. *arXiv preprint arXiv:2607.09999*.

David Pape, Jonathan Evertz, and Lea Schönherr. 2026. The silent hyperparameter: Quantifying the impact of inference backends on llm reproducibility. *arXiv preprint arXiv:2605.19537*.

Hari Prasad and Ritam Pal. 2026. The joint effect of quantization and sampling temperature on llm safety alignment: A factorial analysis. *arXiv preprint arXiv:2606.29581*.

Anton Rasmussen and Hong Qin. 2026. When calibration depends on the scoring rule: Quantized biomedical llm classification. *arXiv preprint arXiv:2608.03854*.

Xuan Ren, Weiqi Zhai, Tianle Pu, Yihua Zhu, Hu Wei, and Bing Zhao. 2026. Right answer, wrong method: Shortcut hacking misleads the evaluation of llm reasoning on frontier science benchmarks. *arXiv preprint arXiv:2608.02442*. Work in progress.

Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. 2023. Are emergent abilities of large language models a mirage? In *Advances in Neural Information Processing Systems (NeurIPS)*.

Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. 2024. Quantifying language models' sensitivity to spurious features in prompt design or: How i learned to start worrying about prompt formatting. In *International Conference on Learning Representations (ICLR)*.

Jonathan von Rad, Yong Cao, and Andreas Geiger. 2026. Unicomp: A unified evaluation of large language model compression via pruning, quantization and distillation. *arXiv preprint arXiv:2602.09130*.

Jiacheng Wang, Yejun Zeng, Jinyang Guo, Yuqing Ma, Aishan Liu, and Xianglong Liu. 2025. Slmquant: Benchmarking small language model quantization for practical deployment. *arXiv preprint arXiv:2511.13023*.

Qingchen Yu, Zifan Zheng, Shichao Song, Zhiyu Li, Feiyu Xiong, Bo Tang, and Ding Chen. 2025. xfinder: Large language models as automated evaluators for reliable evaluation. In *International Conference on Learning Representations (ICLR)*.

## A Frozen configuration

## B Annotation rubric

## C Pre-registration